MAJOR PROJECT STAGE 1

GENAI-ASSISTED AML COMPLIANCE SYSTEM

REVIEW 1

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1. ABSTRACT

Money laundering is a major financial crime that threatens the integrity of financial institutions and the global economy. Conventional Anti-Money Laundering (AML) systems mainly rely on predefined rules, fixed thresholds, and manual investigation. These approaches often generate excessive false positives, require considerable human effort, and may fail to detect complex or previously unseen laundering patterns.

This project proposes a GenAI-Assisted AML Compliance System that combines Machine Learning, Artificial Intelligence, Natural Language Processing, Generative AI, and explainable analytics to strengthen transaction monitoring and suspicious activity detection. The system analyzes transaction amount, frequency, location, account behavior, beneficiary relationships, and transaction history to identify anomalous patterns and assign customer or transaction risk scores. Generative AI assists compliance analysts by providing understandable reasons for alerts, summarizing investigation findings, identifying contributing risk factors, and supporting the preparation of compliance reports.

A secure investigation dashboard enables authorized personnel to review alerts, examine customer and transaction risk information, manage investigation status, and prioritize high-risk cases. The system also incorporates access controls, audit logging, data protection, explainability, and human validation to support responsible AI use and regulatory compliance.

By reducing unnecessary alerts and improving investigation efficiency, the proposed solution provides a scalable and interpretable decision-support system while ensuring that final compliance decisions remain under human oversight.

Keywords: Anti-Money Laundering, Transaction Monitoring, Anomaly Detection, Risk Scoring, Generative AI, Explainable AI, Suspicious Activity Report, Compliance Dashboard.

2. INTRODUCTION

2.1 Background

Money laundering is the process of disguising the proceeds of crime so that they appear to originate from legitimate sources. It typically passes through three stages: placement (introducing illicit cash into the financial system), layering (moving funds through many accounts and transactions to obscure their origin) and integration (re-introducing the funds into the economy as apparently legitimate wealth). Banks, payment providers and other regulated entities are legally required to monitor customer activity and report suspicious transactions to the financial intelligence authority of their country, in line with international standards such as the FATF Recommendations. In India, these obligations arise mainly from the Prevention of

Money Laundering Act (PMLA), 2002 and the guidelines issued by the Reserve Bank of India and the Financial Intelligence Unit–India (FIU-IND).

Most institutions still depend on rule-based monitoring, in which a transaction is flagged when it crosses a fixed threshold or matches a predefined scenario. Because criminals continuously adapt their behaviour to stay below such thresholds, rule-based systems produce large numbers of false alerts while still missing new typologies. Analysts then spend most of their time reviewing harmless alerts and writing narrative reports manually.

2.2 Motivation

Recent progress in machine learning allows behaviour-based detection of unusual activity, and advances in Generative AI and Natural Language Processing make it possible to explain model outputs and draft investigation narratives in plain language. However, a risk score alone is not sufficient in a regulated environment: analysts and auditors must understand why an alert was raised and must remain accountable for the final decision. This project is motivated by the need for an AML solution that is accurate, explainable, auditable and designed around human oversight.

2.3 Objectives

- To design a transaction-monitoring system that detects anomalous and potentially suspicious activity using machine learning.
- To assign interpretable risk scores to customers and transactions using amount, frequency, location, account behaviour, beneficiary relationships and transaction history.
- To reduce false-positive alerts and help analysts prioritise high-risk cases.
- To use Generative AI for alert explanations, investigation summaries, identification of risk factors and drafting of compliance reports.
- To provide a secure investigation dashboard with role-based access, case management and audit logging.
- To keep a human analyst in the loop so that every final compliance decision is validated by an authorised person.

2.4 Scope

The system is a decision-support tool for AML compliance teams. It covers data ingestion, feature engineering, anomaly detection, risk scoring, explainability, GenAI-assisted summarisation and report drafting, and an investigation dashboard. The prototype is developed and evaluated on synthetic or anonymised transaction data; it does not automatically block

accounts or file reports with regulators without human approval. Integration with live core-banking systems is considered future work.

3. LITERATURE REVIEW

Research on AML technology has moved from fixed rules to statistical learning, graph analysis, explainable models and, most recently, large language models. The table below summarises the main directions that inform this project.

Approach	Representative work	Strengths	Limitations
Rule / threshold-based monitoring	FATF Recommendations; typology-based scenario rules used in commercial AML tools	Simple, transparent and easy to justify to regulators	High false-positive rate; static; misses new patterns; needs frequent manual tuning
Supervised machine learning	Gradient boosting (Chen & Guestrin, XGBoost, 2016); Random Forest	Strong accuracy when labelled data is available	Confirmed laundering labels are rare and imbalanced; poor on unseen typologies
Unsupervised anomaly detection	Isolation Forest (Liu, Ting & Zhou, 2008); autoencoders	No labels needed; finds unusual behaviour	Anomalous does not always mean suspicious; scores are hard to explain
Graph-based learning	Weber et al., "Anti-Money Laundering in Bitcoin" using Graph Convolutional Networks (2019); Altman et al., synthetic AML transaction datasets (NeurIPS 2023)	Captures networks of accounts and layering patterns	Computationally heavy; needs relationship data; limited interpretability
Explainable AI	SHAP (Lundberg & Lee, 2017); LIME	Shows which features drove each score; supports audit	Outputs are technical and not directly readable by non-technical staff
Large language models / GenAI	Large pre-trained language models (e.g., Brown et al., 2020)	Produces fluent explanations, summaries and report drafts	Can hallucinate; needs grounding in verified data and human review

3.1 Research Gap

Existing literature treats detection, explanation and reporting largely as separate problems. Machine-learning studies focus on improving detection metrics but give analysts little help in understanding alerts, while language-model studies seldom connect to a real alert and case-management workflow with access control and audit trails. There is therefore a need for an integrated system in which (i) a machine-learning engine detects and scores risk, (ii) explainability methods identify the contributing factors, (iii) Generative AI converts those

factors into understandable explanations and report drafts grounded in the underlying data, and (iv) humans validate every final decision. This project addresses that gap.

Note: *references listed above should be verified and formatted in the department-prescribed style (IEEE/APA) in the final report.*

4. PROBLEM STATEMENT

Financial institutions process millions of transactions every day, and regulators expect suspicious activity to be identified and reported promptly. Traditional AML monitoring based on fixed rules and thresholds creates too many false positives, depends heavily on manual investigation, and cannot reliably detect complex or previously unseen laundering patterns. Even where machine-learning models are used, their outputs are often opaque, and analysts must still interpret scores and write case narratives by hand.

4.1 Key Problems Identified

- **Excessive false positives:** the large majority of alerts generated by rule-based systems are not genuine suspicious cases, wasting investigator time.
- **Manual effort:** reviewing alerts, collecting customer and transaction history, and writing compliance reports are slow, repetitive tasks.
- **Missed complex patterns:** structuring, layering through multiple accounts, and new typologies are difficult to capture with static rules.
- **Lack of explainability:** analysts and auditors cannot easily see why a customer or transaction was flagged.
- **Weak prioritisation:** alerts are often handled in arrival order instead of by risk, delaying high-risk cases.
- **Governance and security concerns:** sensitive financial data requires access control, audit logging, and responsible use of AI with human oversight.

4.2 Problem Definition

To design and develop an explainable, secure and scalable decision-support system that analyses transaction amount, frequency, location, account behaviour, beneficiary relationships and transaction history to detect anomalous activity, assign risk scores, reduce unnecessary alerts, and use Generative AI to explain alerts and assist report preparation, while keeping final compliance decisions under human control.

5. EXISTING SYSTEM

In most banks and financial institutions today, AML compliance is performed by a transaction-monitoring system built around predefined rules and fixed thresholds, followed by manual investigation by compliance analysts.

5.1 Working of the Existing System

1. Transactions are collected from core banking and payment channels.
2. Static rules are applied, for example: cash deposits above a set limit, many transactions just under a reporting threshold, rapid in-and-out movement of funds, or transfers to high-risk jurisdictions.
3. Every transaction matching a rule generates an alert, regardless of customer context.
4. Analysts manually open each alert, gather customer profile and past transaction history from several systems, and judge whether it is suspicious.
5. For suspicious cases, the analyst writes the narrative manually and prepares a Suspicious Transaction/Activity Report for the compliance officer.

5.2 Limitations of the Existing System

- Fixed thresholds can be learnt and avoided by criminals, and are not adapted to individual customer behaviour.
- A very high proportion of alerts are false positives, causing alert fatigue and backlog.
- Unable to detect new or complex laundering patterns that do not match any written rule.
- Little or no explanation of why an alert was raised beyond the name of the rule.
- Information is scattered across systems, making investigation slow and inconsistent.
- Report writing is manual, time-consuming and varies from analyst to analyst.
- Rule maintenance requires continuous manual tuning and does not scale with growing transaction volumes.

6. PROPOSED SYSTEM

6.1 Overview

The proposed GenAI-Assisted AML Compliance System combines machine learning, explainable analytics and Generative AI in a single workflow. Transaction and customer data are cleaned and converted into behavioural features; machine-learning models detect anomalies and assign risk scores; explainability techniques identify the factors responsible; and a Generative AI component turns these factors into clear explanations, case summaries and draft

compliance reports. Authorised analysts review everything on a secure dashboard and make the final decision.

6.2 Key Features

- **Behaviour-based detection:** analyses amount, frequency, location, account behaviour, beneficiary relationships and transaction history instead of only fixed thresholds.
- **Customer and transaction risk scoring:** every transaction and customer receives a risk score so that high-risk cases are prioritised.
- **Explainable alerts:** contributing risk factors are listed for each alert (for example, using SHAP values).
- **GenAI assistance:** plain-language alert reasons, investigation summaries and drafts for compliance reports, grounded only in the case data.
- **Secure investigation dashboard:** alert queue, customer and transaction risk views, case status management and priority ranking.
- **Responsible AI controls:** role-based access control, audit logging, data protection, and mandatory human validation of decisions.

6.3 Workflow of the Proposed System

1. Ingest transaction, customer and beneficiary data and validate it.
2. Pre-process the data and engineer behavioural features (e.g., transaction velocity, deviation from customer average, new-beneficiary flag, geographic risk).
3. Run the machine-learning anomaly-detection model and compute risk scores.
4. Generate alerts for cases above the risk threshold and compute contributing factors.
5. Use Generative AI to write the alert reason and case summary from the verified factors.
6. Analyst reviews the alert on the dashboard, adds notes and updates the investigation status.
7. If suspicious, a draft compliance report is generated, edited and approved by the compliance officer.
8. All actions are written to the audit log; analyst feedback is used to improve the model and reduce false positives.

6.4 Comparison: Existing vs Proposed System

Aspect	Existing System	Proposed System
Detection method	Fixed rules and thresholds	Machine-learning behaviour analysis plus risk scoring

Aspect	Existing System	Proposed System
False positives	High	Reduced through risk-based alerting and feedback
New patterns	Mostly missed	Anomaly detection can flag unseen patterns
Explainability	Only rule name	Contributing factors and GenAI plain-language reasons
Reporting	Fully manual	GenAI-drafted summaries and reports, human-approved
Prioritisation	Arrival order	Ranked by risk score
Governance	Limited traceability	RBAC, audit logs and human-in-the-loop validation

6.5 Advantages

- Fewer unnecessary alerts and faster investigations.
- Better detection of complex and evolving laundering patterns.
- Transparent, interpretable and auditable decision support.
- Scalable to large transaction volumes.
- Final decisions remain with human analysts, supporting regulatory acceptance.

7. HARDWARE REQUIREMENTS

The following hardware is recommended for developing, training and running the prototype. Production deployment would be sized according to transaction volume.

7.1 Development / Client System

Component	Minimum / Recommended Specification
Processor	Intel Core i5 (8th gen) or AMD Ryzen 5 or higher
RAM	8 GB minimum; 16 GB recommended for model training
Storage	256 GB SSD minimum; 512 GB SSD recommended
GPU (optional)	NVIDIA GPU with 4 GB+ VRAM for faster model training / local LLM inference
Display	1366 x 768 minimum; 1920 x 1080 recommended
Network	Broadband internet connection (for package installation and LLM API access)
Input devices	Standard keyboard and mouse

7.2 Server / Deployment System (Prototype)

Component	Specification
Processor	4 vCPUs or more (Intel Xeon / AMD EPYC class)
RAM	16 GB or more
Storage	500 GB SSD with regular backup
GPU (optional)	NVIDIA T4 class or better if a local open-source LLM is hosted
Network	Secure, encrypted connectivity (HTTPS/TLS) with firewall protection